

crypto-frameworks

Crypto Compliance Framework Mapping

This document maps each Preston-Check crypto check (P-301..P-360) to the authoritative frameworks it satisfies. It serves two purposes: it lets operators run framework-scoped scans (`preston-check --framework MiCA`), and it gives auditors a clear evidence map from check findings to the specific framework controls they cover.

The runner's `--framework` flag does a case-insensitive substring match against each check's `frameworks` metadata field. Framework names below match the canonical short codes used in metadata.

OWASP Smart Contract Top 10 (2025)

Filter: `--framework "OWASP-SC-Top-10:2025"`

CHECK	SUB-CATEGORY
P-301 Reentrancy	SC05
P-302 Integer Overflow	SC03
P-303 tx.origin Authorization	SC02
P-304 Unbounded Loops / Gas DoS	SC09
P-305 DEX Slippage Protection	SC04
P-306 Oracle Manipulation	SC07
P-307 Unbounded Token Approvals	SC04
P-308 Bridge Replay Protection	SC02
P-310 Governance Time-Locks	SC01

CHECK	SUB-CATEGORY
P-348 Proxy Upgrade Safety	SC01
P-349 EIP-2612 Permit Replay	SC02
P-350 ERC-4337 Account Abstraction Safety	SC02
P-351 Access Control (Comprehensive)	SC01
P-352 Flash Loan Attack Resistance	SC04
P-353 Unchecked External Calls	SC07
P-356 Liquidation Safety	SC02
P-357 Token Admin Freeze Handling	SC04

CryptoCurrency Security Standard v9.0 (CCSS)

Filter: `--framework CCSS` for all CCSS-mapped checks; or `--framework "CCSS:9.0:Level1"` / `Level2` / `Level3` for tier-specific.

CHECK	CCSS LEVEL
P-311 Private Key / Mnemonic Exposure	Level 1
P-312 HSM Integration	Level 2
P-313 KMS-Only Key Operations	Level 2
P-314 MPC / Threshold Signatures	Level 2
P-315 Cold Wallet Air-Gap	Level 3
P-316 Hot Wallet Concentration Limits	Level 2
P-317 Key Ceremony Documentation	Level 1+
P-318 Seed Phrase Handling Hygiene	Level 1
P-319 Key Rotation Schedule	Level 2
P-320 Multi-Sig Approval Discipline	Level 2

CHECK	CCSS LEVEL
P-344 Wallet Backup Encryption	Level 1
P-345 Shamir’s Secret Sharing	Level 3
P-354 Cold-to-Hot Transfer Verification	Level 3
P-358 CCSS Compliance Evidence	Level 1+

FATF Recommendations and Travel Rule

Filter: ---framework FATF

CHECK	FATF RECOMMENDATION
P-322 Withdrawal Whitelist	Rec.16
P-323 Pre-Send Risk Scoring	Rec.16
P-324 OFAC SDN Live Screening	Rec.6
P-327 Mixer Detection	Rec.16
P-328 Travel Rule Compliance	Rec.16
P-335 Counterparty Reputation	Rec.10
P-336 Tainted Funds Source Trace	Rec.10

OFAC and Sanctions

Filter: ---framework OFAC

CHECK	MAPPING
P-323 Pre-Send Risk Scoring	OFAC:2024
P-324 OFAC SDN Live Screening	OFAC:2024
P-327 Mixer Detection	OFAC:2024 (mixer-sanctioning precedent)
P-328 Travel Rule	OFAC:2024

CHECK	MAPPING
P-336 Tainted Funds Trace	OFAC:2024

EU MiCA / TFR

Filter: --framework MiCA or --framework EU-TFR

CHECK	MAPPING
P-316 Hot Wallet Concentration	MiCA:2024 (segregation, secure cold storage)
P-320 Multi-Sig Discipline	MiCA:2024 (authorization controls)
P-322 Withdrawal Whitelist	MiCA:2024 (custody segregation)
P-328 Travel Rule	EU-TFR:2023 (Transfer of Funds Regulation)
P-358 CCSS Evidence	MiCA:2024 (custody audit evidence)

EU MiCA itself requires segregation, secure cold storage, multi-signature wallets, strong key management, daily reconciliation, and regular audits; the checks above touch each of those dimensions.

NIST Cybersecurity Framework 2.0

Filter: --framework NIST-CSF

CHECK	NIST-CSF FUNCTION:CATEGORY
P-312 HSM Integration	PR.DS
P-313 KMS-Only Operations	PR.DS
P-314 MPC / TSS	PR.DS
P-315 Cold Wallet Air-Gap	PR.DS
P-316 Hot Wallet Caps	PR.DS, DE.AE
P-317 Key Ceremony	GV.RM
P-319 Key Rotation	PR.DS

CHECK	NIST-CSF FUNCTION:CATEGORY
P-320 Multi-Sig	PR.AC
P-321 Address Validation	PR.DS
P-322 Withdrawal Whitelist	PR.AC
P-329 New-Address Cooldown	PR.AC
P-330 Step-Up Auth	PR.AC
P-331 Dust Attack Detection	DE.CM
P-335 Counterparty Reputation	DE.AE
P-337 Compromise Indicators	DE.AE
P-338 Approval Audit	PR.AC
P-339 Drainer Pattern Detection	DE.CM
P-340 Reorg Confirmation Depth	PR.DS
P-341 Hardware Wallet	PR.DS
P-342 Secure Enclave / TPM	PR.DS
P-343 BIP-44 Derivation	PR.DS
P-344 Backup Encryption	PR.DS
P-345 Shamir SSS	PR.DS
P-346 Memory Hygiene	PR.DS
P-354 Cold-to-Hot Verification	PR.AC
P-355 Blind Signing Prevention	PR.AC
P-359 Pause / Circuit Breaker	RS.MI
P-360 PQC Readiness	PR.DS

NIST Post-Quantum Cryptography (FIPS)

203/204/205)

Filter: --framework FIPS or --framework "NIST-FIPS"

CHECK	MAPPING
P-360 PQC Readiness Assessment	NIST-FIPS:203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA)

OWASP Mobile Application Security Verification Standard 2.0

Filter: ---framework OWASP-MASVS

CHECK	MASVS CONTROL
P-342 Secure Enclave / TPM	CRYPTO-1
P-344 Wallet Backup Encryption	CRYPTO-2
P-355 Blind Signing Prevention	AUTH-2

OWASP API Security Top 10 (2023)

Filter: ---framework OWASP-API

CHECK	API TOP 10 ITEM
P-311 Private Key Exposure	API8
P-318 Seed Phrase Handling	API8
P-321 Address Validation	API3
P-326 Address Poisoning Detection	API3
P-338 Approval Audit	API3

ISO 27001:2022

Filter: ---framework ISO-27001

CHECK	ANNEX A CONTROL
P-312 HSM Integration	A.10.1
P-313 KMS-Only Operations	A.10.1
P-315 Cold Wallet Air-Gap	A.10.1
P-318 Seed Phrase Handling	A.8.10
P-319 Key Rotation	A.10.1
P-320 Multi-Sig	A.5.16
P-322 Withdrawal Whitelist	A.5.16
P-329 New-Address Cooldown	A.5.16
P-330 Step-Up Auth	A.5.17

PCI-DSS v4.0

Filter: ---framework PCI-DSS

CHECK	PCI REQUIREMENT
P-311 Private Key Exposure	3.5
P-318 Seed Phrase Handling	3.5
P-313 KMS-Only Operations	3.6
P-319 Key Rotation	3.6
P-344 Wallet Backup Encryption	3.5

SOC 2 Trust Service Criteria (2017)

Filter: ---framework SOC2

CHECK	TSC
P-317 Key Ceremony	CC6.1

PSD2 (EU Payment Services Directive 2)

Filter: --framework PSD2

CHECK	ARTICLE
P-330 Step-Up Auth	Art.97 (SCA)

FinCEN

Filter: --framework FinCEN

CHECK	CITATION
P-323 Pre-Send Risk Scoring	31CFR1010
P-328 Travel Rule	31CFR1010.410
P-336 Tainted Funds Trace	31CFR1010

NIST Secure Software Development Framework

Filter: --framework NIST-SSDF

CHECK	SSDF PRACTICE
P-301 Reentrancy	PW.7

Compliance run-recipes

For an EU MiCA-focused custody audit dry-run:

```
preston-check --framework MiCA --report mica-audit.md --config configs/myapp.yml
```

For CCSS Level 2 self-assessment:

```
preston-check --framework "CCSS:9.0:Level2" --report ccss-l2.md
```

For an OWASP Smart Contract Top 10 (2025) review:

```
preston-check --framework "OWASP-SC-Top-10:2025" --report owasp-sc.md
```

For comprehensive sanctions and Travel Rule compliance:

```
preston-check --framework FATF --report fatf-compliance.md  
preston-check --framework OFAC --report ofac-compliance.md
```

Note that filters are case-insensitive substrings against the `frameworks` metadata field. A filter like `MiCA` matches any check whose metadata mentions `MiCA:2024` or future versions; a filter like `CCSS:9.0:Level2` matches only checks pinned to that specific level. This means the same operator command continues to work as new framework versions land — the catalog stays scoped to the latest references the checks declare.